INCH-POUND
MIL-DTL-24643/30G
27 November 2023
SUPERSEDING
MIL-DTL-24643/30F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 600 VOLTS, TYPE LS1SWU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Watertight)

First	-	Copper conductor, coated or uncoated, ASTM B286, size 22 through size 7 strands.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation, white or natural. Nominal wall thickness of 0.048 inch. Wall thickness may vary from nominal as necessary in order for completed cable to meet the specified electrical requirements.
Fourth	-	Braided shield of 36 AWG, coated or uncoated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.
Fifth	-	Shield insulation of two polyester tapes, outer tape sealed. An alternate shield insulation may consist of one polyester tape, plus an extruded jacket of a suitable transparent material; minimum average thickness of the transparent material 0.003 inch. Standard identification code shall be applied by method 2.
Sixth	-	The specified number of shielded singles (see table I) shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers may be used as necessary to form a firm, well-rounded assembly.
Seventh	-	An optional binder.
Eighth	-	Cross-linked polyolefin jacket. See table I for thickness.

NOTE: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

Distribution Statement A. Approved for public release: distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:05Z
Check the source to verify that this is the current version before use.

TABLE I. Details.

Military part no. M24643/30	Type and size	Number of conductors	Cable jacket thickness min. avg. (inch)	Overall diameter (inches)		Hydrostatic leakage (cubic inches)
				min.	max.	
-01UN	LS1SWU-2	2	0.038	0.430	0.455	1.00
-02UN	LS1SWU-14	14	0.050	0.825	0.870	1.00
-03UN	LS1SWU-20	20	0.050	0.970	1.030	1.50
-04UN	LS1SWU-30	30	0.063	1.135	1.200	2.00

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	17.2
Voltage withstand (volts, root mean square, min.)	
Conductor to component shield	3,000
Component shield to component shield	500
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to component shield	500
Component shield to component shield	100
Conductor continuity	No failure
Shield continuity	No failure

Group A:

Visual and dimensional	No failure
Hydrostatic (open end) (leakage at 300 lb/in^2, 2 hours, in^3, max.)	See table I
Mutual capacitance (nominal at 1 kilohertz [kHz]) (picofarad [pF]/100 feet)	28
Characteristic impedance, at 1 megahertz (ohms)	61±3

Group B:

Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:05Z
Check the source to verify that this is the current version before use.

Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Drip (203±1.8 °F [95±1 °C])	No failure

Group C:

Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure
Shield (conformance to material, construction, and coverage)	No failure

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:05Z
Check the source to verify that this is the current version before use.

Shield insulation	2
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure
Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LS1SWU all sizes	1,000

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:05Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-046)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://asssist.dla.mil.

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T00:05Z
Check the source to verify that this is the current version before use.